

Assessing Power Usage Effectiveness in Serverless Computing Environments

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- As the computational power of machines, servers, and clusters has not ceased to grow, the energy consumed to use, cool, and maintain these computational resources has also grown, generating a huge ecological crisis in the domain.
 - The energy consumed by the data-centers themselves has already achieved 1% of the global energy consumed, and may continue to increase [1].
- Serverless computing has the potential of saving resources by reusing previous computational environments, and by releasing resources as soon as functions are executed, it is a promising paradigm to be used towards a greener computing [2, 3, 4].
 - However, the efforts made in this direction all rely on power usage measurements and energy consumption estimations.
 - To the best of our knowledge, no accurate metrics have been explored in the context of serverless computing.

We propose a preliminary exploration of the *Power Usage Effectiveness* (PUE) of serverless. Our main contributions are:

- We quantify energy overheads in serverless executions;
- We propose a reproducible setup [5];
- We demonstrate how POWERAPI [6] provides fine-grained energy measurements in a serverless environment;
- We adapt the suite of benchmarks named FUNCTIONBENCH [7];
- We propose a first approach on how to compute the named vPUE and cPUE of serverless functions;
- Finally, we detail our exploration of the different relations between the PUE of serverless functions and
 - their starting mode;
 - the size of serverless platforms;
 - the loading of serverless functions.

Power Usage Effectiveness:

- The *Power Usage Effectiveness* (PUE) can be defined as the ratio of the total energy consumed at the data-center level to the energy consumed by hosted IT equipment.
 - This metric is considered optimal when it is lowered to 1.
- More specifically, we consider extended versions of the PUE: the *virtual PUE* (vPUE) and the *combined PUE* (cPUE) of serverless applications [8, 9].
 - The vPUE computes the PUE of individual software layers, which can be stacked indefinitely and accounts for nested virtualization practices,
 - The cPUE reflects the end-to-end efficiency of a software platform.

Preliminary Concepts

Serverless Computing:

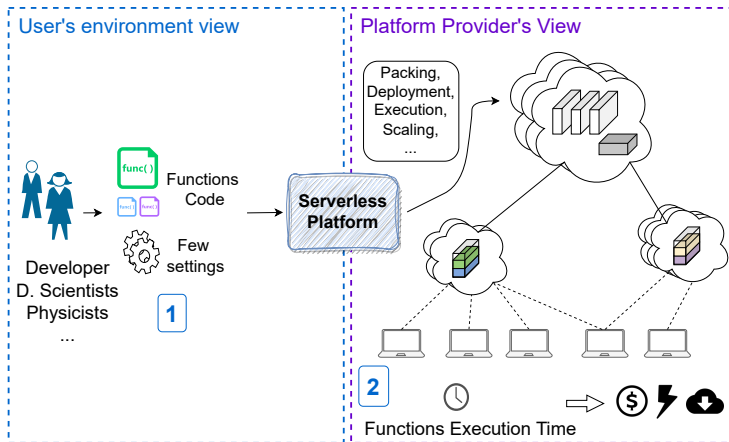


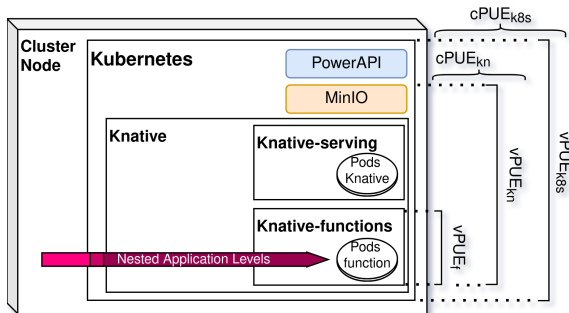
Figure: 1. Serverless computing illustration and definitions [10]

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Serverless vPUE and cPUE

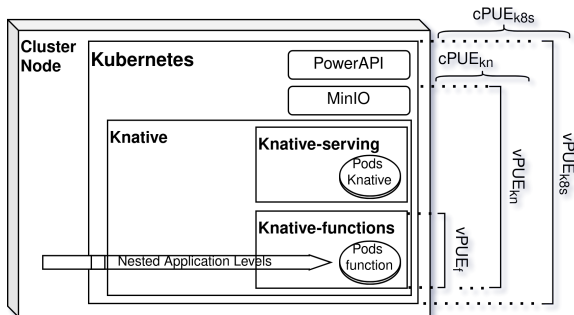
As defined in [8], the vPUE and the cPUE are the ratios between the energy (in Joules in our case) of different components that make up our serverless layers and the energy consumed by our infrastructure. We compute them as illustrated in Figure 8, with the following the equations:



Reproducible artifacts

Figure: 2. Nested Application Levels and the PUE's computation.

Serverless vPUE and cPUE



$$\text{Metric} = \frac{\text{Energy(A)}}{\text{Energy(B)}}$$

$$\text{Metric} = \text{Energy(A)} \times \text{Energy(B)}$$

Figure: 2. Nested Application Levels and the PUE's computation.

$$function_performance = \frac{time(function_execution)}{time(pod_execution)} \quad (1)$$

Serverless vPUE and cPUE

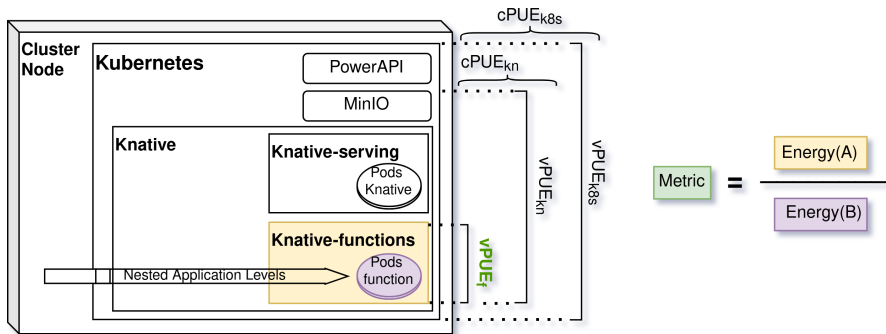


Figure: 2. Nested Application Levels and the PUE's computation.

$$vPUE_f = \frac{\sum \text{energy}(\text{pod})}{\sum \text{energy}(\text{pod}) \times \text{function_performance}} \quad (2)$$

Serverless vPUE and cPUE

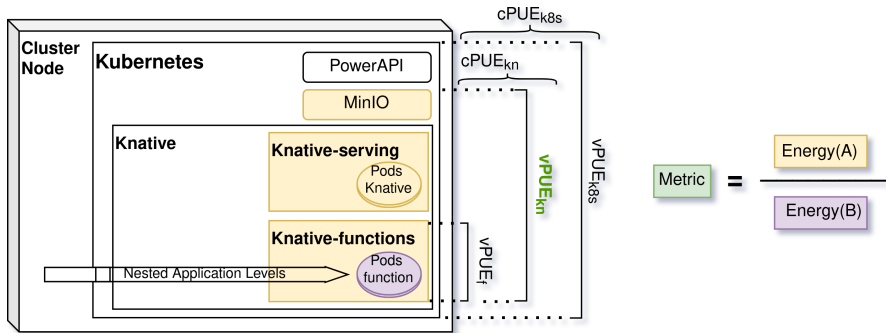


Figure: 2. Nested Application Levels and the PUE's computation.

$$vPUE_{kn} = \frac{\sum energy(Knative) + \sum energy(MinIO) + \sum energy(pod)}{\sum energy(pod)} \quad (3)$$

Serverless vPUE and cPUE

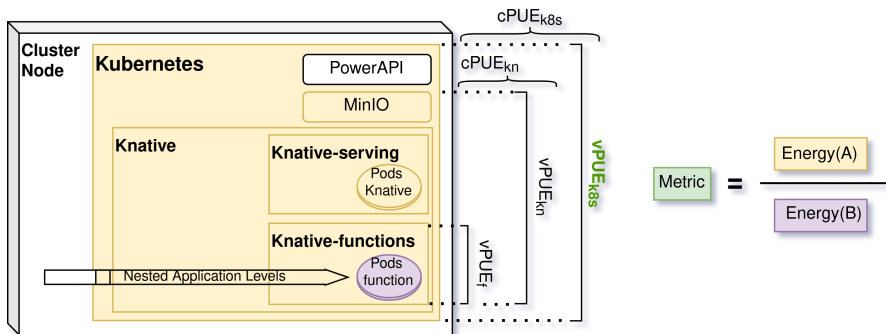


Figure: 2. Nested Application Levels and the PUE's computation.

$$vPUE_{k8s} = \frac{\sum energy(node) - \sum energy(PowerAPI)}{\sum energy(pod)} \quad (4)$$

Serverless vPUE and cPUE

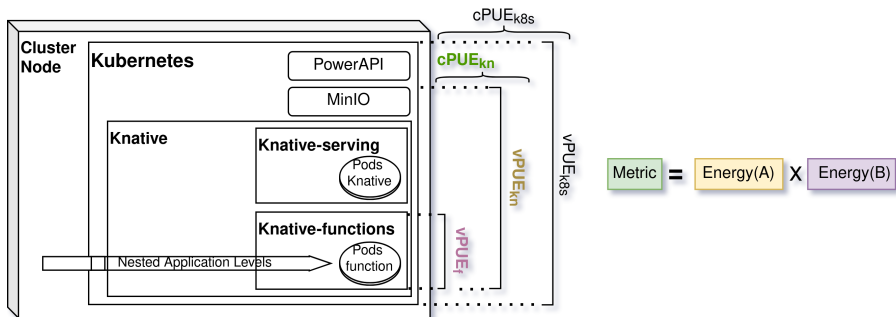


Figure: 2. Nested Application Levels and the PUE's computation.

$$cPUE_{kn} = vPUE_f \times vPUE_{kn} \quad (5)$$

Serverless vPUE and cPUE

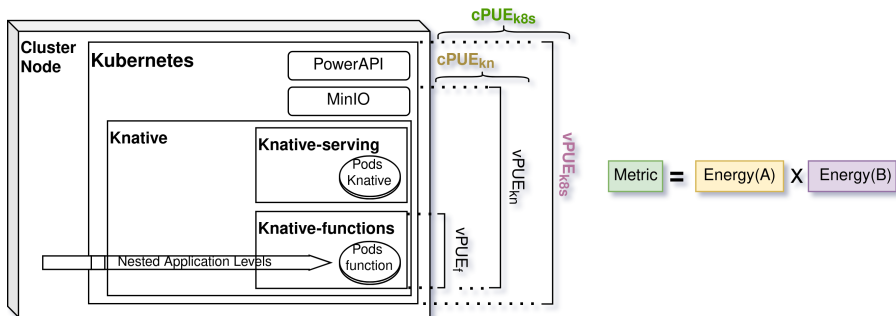


Figure: 2. Nested Application Levels and the PUE's computation.

$$cPUE_{k8s} = cPUE_{kn} \times vPUE_{k8s} \quad (6)$$

Serverless vPUE and cPUE

As defined in [8], the vPUE and the cPUE are the ratios between the energy (in Joules in our case) of different components that make up our serverless layers and the energy consumed by our infrastructure. We compute them as illustrated in Figure 8, following the equations below:

$$function_performance = \frac{time(function_execution)}{time(pod_execution)} \quad (7)$$

$$vPUE_f = \frac{\sum energy(pod)}{\sum energy(pod) \times function_performance} \quad (8)$$

$$vPUE_{kn} = \frac{\sum energy(Knative) + \sum energy(MinIO) + \sum energy(pod)}{\sum energy(pod)} \quad (9)$$

$$vPUE_{k8s} = \frac{\sum energy(node) - \sum energy(PowerAPI)}{\sum energy(pod)} \quad (10)$$

$$cPUE_{kn} = vPUE_f \times vPUE_{kn} \quad (11)$$

$$cPUE_{k8s} = cPUE_{kn} \times vPUE_{k8s} \quad (12)$$

Design of Experiments

Parameters	Batch 1	Batch 2
Workload Size	24 (4 functions $\times$ 2 inputs $\times$ 3 repetitions)	200 (2 functions $\times$ 100 repetitions)
Platform Size	1 control plane + 1 worker	1 control plane + 1, 2, 3 workers
Execution Mode	Cold, Warm and Hot	Hot
Concurrency Target	70%	50, 70, 100%
Total of executions	72	3,600

Table: 1. Design of Experiments. Total of 3672 experiments.

We evaluated 5 functions with different inputs. In total, we cover 10 combinations of functions and inputs, from FunctionBench [7, 11, 12]:

- Linpack (35000 and 5000 nxn matrices);
- Chameleon (4000 and 8000 nxn matrices);
- Matrix Multiplication (25000 and 40000 nxn matrices);
- Model Training (50 and 100 Mb datasets);
- Video Processing (50 and 100 Mb video sizes).

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Serverless executions & Challenges

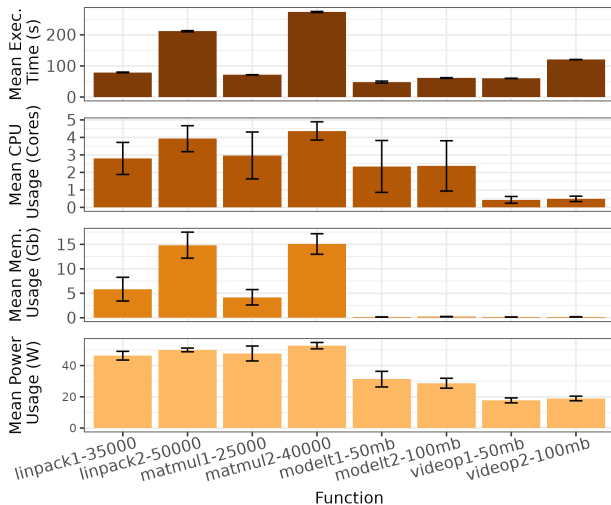


Figure: 3. Execution time and resource usage of the serverless functions

Performance over the functions execution mode - vPUE

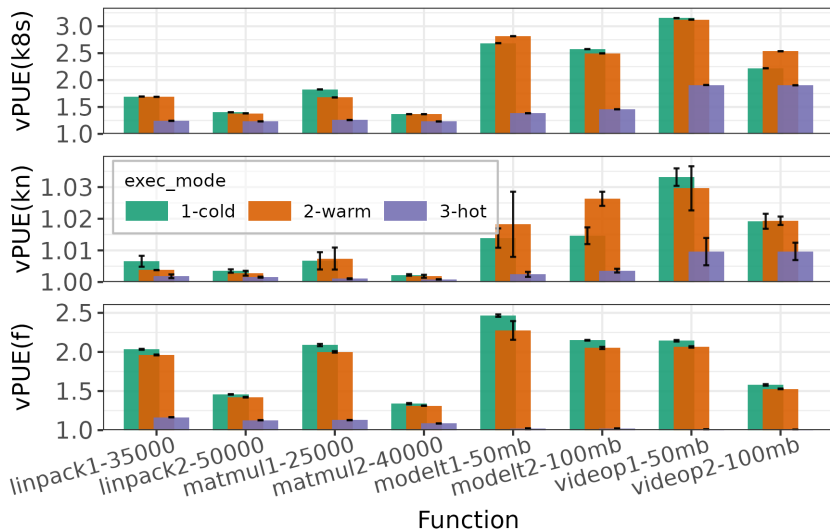


Figure: 4. vPUE for different parameters & functions.

Performance over the functions execution mode - cPUE

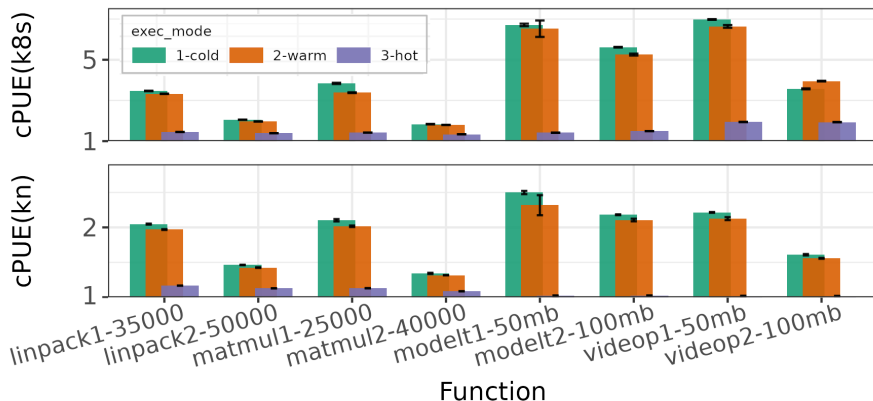


Figure: 5. cPUE for different parameters & functions.

Overall Serverless cPUE

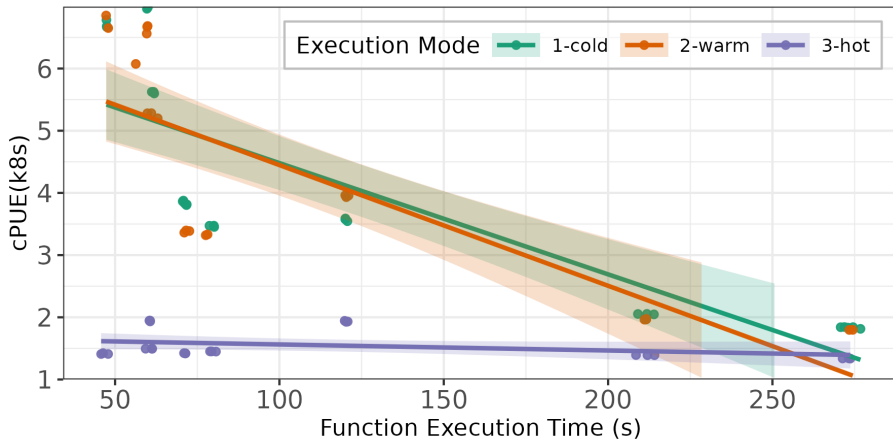


Figure: 6. Linear regression of function execution time over the cPUE.

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Our findings acknowledge that

- The starting mode of functions in a serverless platform is critical.
- Hot start-ups are really beneficial in terms of the overall PUE.
- Even as serverless was designed for very fast execution function (from milliseconds to a few minutes), the first part of this range is not a good approach in terms of PUE.
 - In fact, as long as we use the deployed functions for a longer duration, their PUE is improving, achieving close to optimal performance.
 - On the contrary, our results showed an increase of by $2.5\times$ of $vPUE_f$ for cold and warm-started functions, in comparison to hot ones; arriving at by $7\times$ worst performance in a global view within $cPUE_{k8s}$.

We plan to expand our experimentation setup:

- To include a wider range of serverless functions, also including workflow executions;
- To add more PUE metrics to improve our exploration campaign, such as the *partial PUE* (pPUE) to better model the PUE of external but included services, such as *MinIO*.

Acknowledgments

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Serverless executions & Challenges

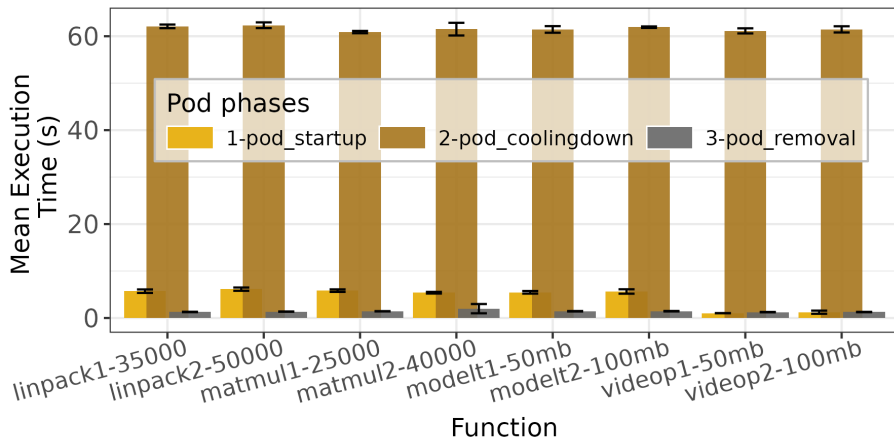


Figure: 7. Mean duration of different phases of pods execution.

Impacts on number of nodes

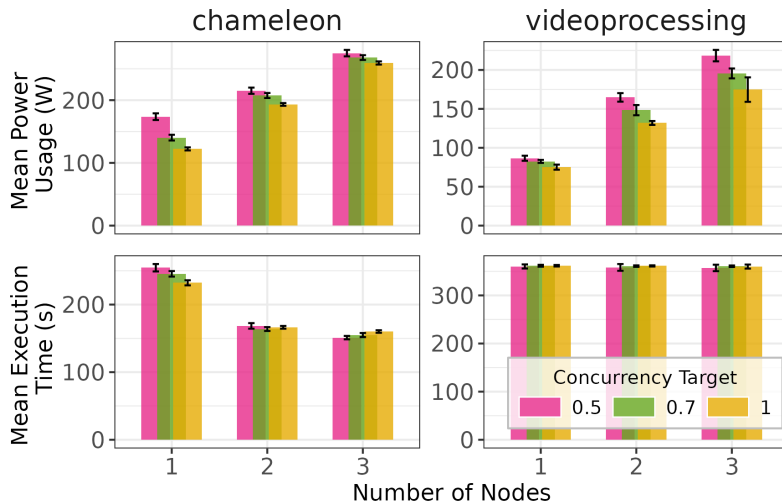


Figure: 8. The effects of concurrency auto-scaling & number of nodes.